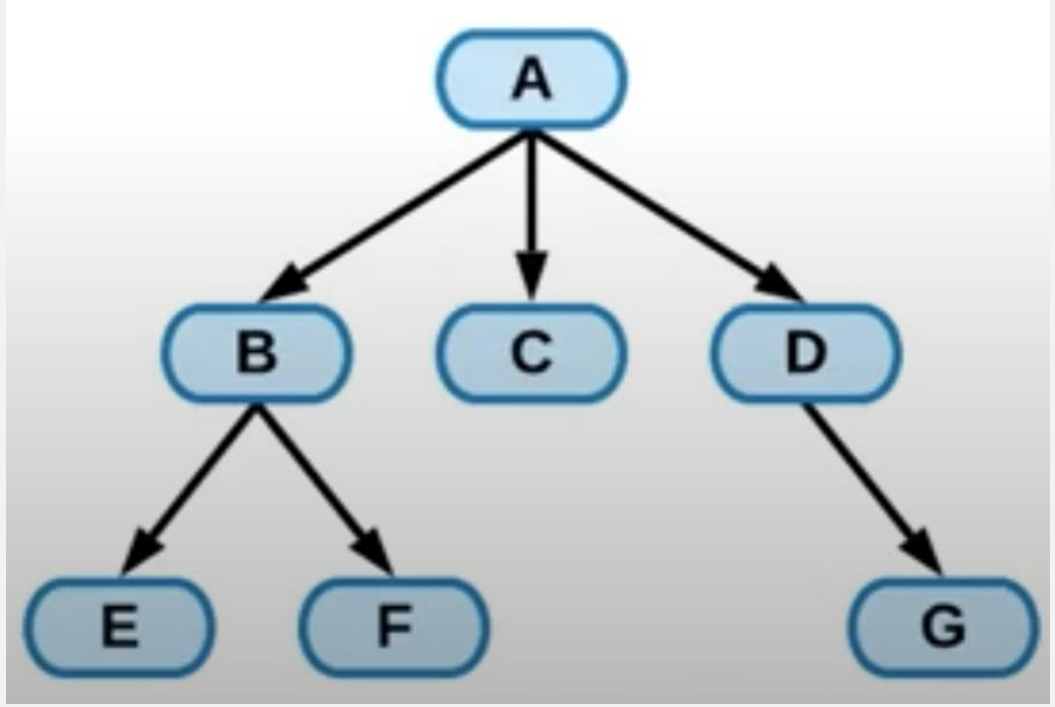
Graphics Programming

Framework Class

GB. Tony Liang

Trees and Scene Graphs

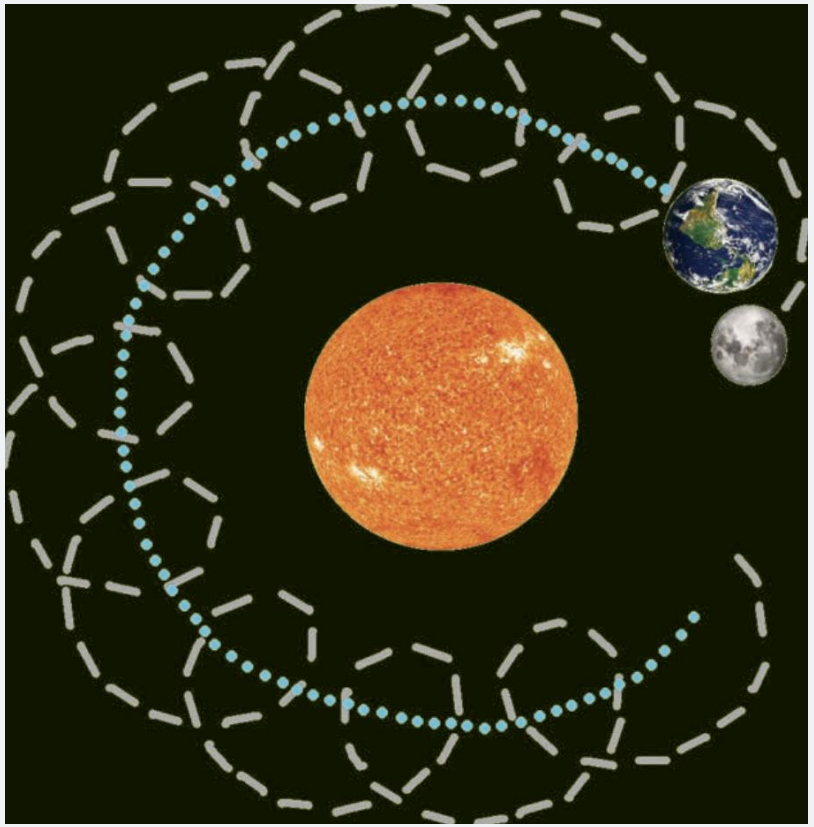


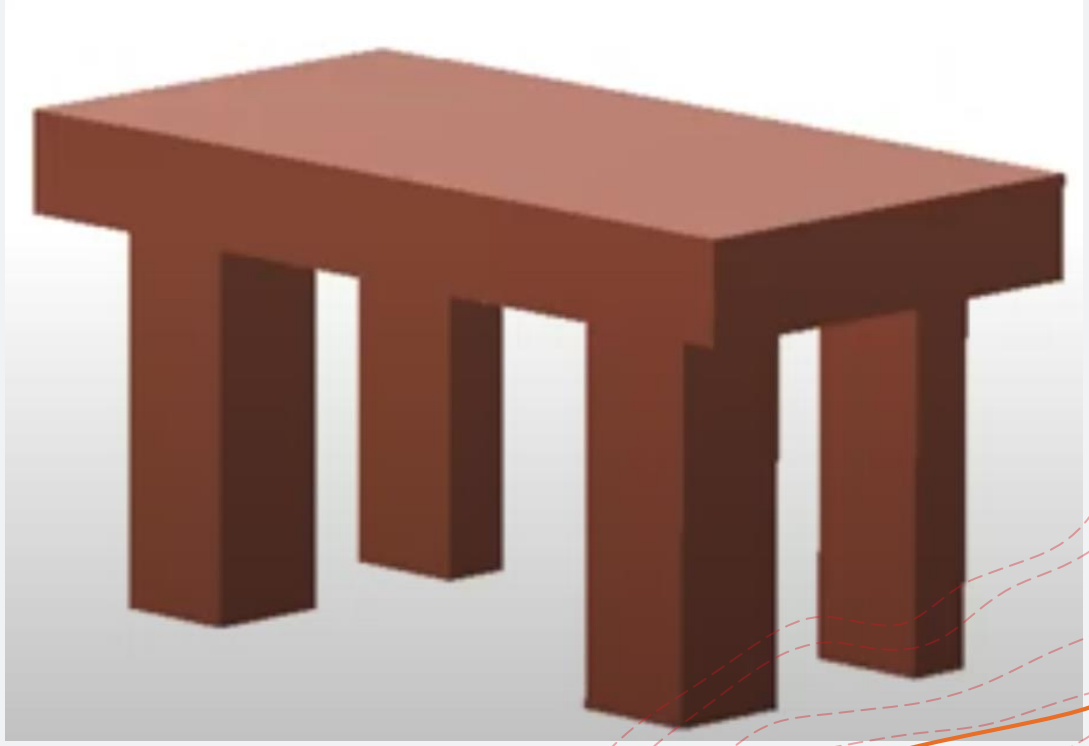
- Scene Graph
  - Organizes contents of a 3D scene
  - Using a hierarchical or tree-like structure
- Tree
  - Collection of *node* objects
- Node
  - Stores value/object, list of *child* nodes, reference to *parent*
- Addition terms
  - *Root* node, *descendants/ancestors* of a node

Why Scene Graph

+Simplifies relative motion

+Enable reuse of objects





Object3D

- Object3D
  - Scene graph nodes, represent objects in world
    - A matrix to store its local transformation
    - A list of reference to child objects
    - A reference to a parent object
- Object 3D class extensions
  - Scene class – root node
  - Group class – interior node
  - Mesh class – visible objects to render
  - Camera class – represents viewer
  - Light class – affects shading and shadows

Mesh, Geometry, Material

- For maximum flexibility, Mesh class stores
  - Transformation matrix
  - Vertex Array Object
  - Instances of Geometry() and Material() classes
- Geometry class
  - Specify general shape, other vertex-related properties
  - Store Attribute objects
- Material class
  - Specific the general appearance of an object
  - Stores shader code, Uniform objects, render settings

Renderer

- Handles general OpenGL initialization tasks and rendering the image
- Render function requires scene and camera parameters
- For each mesh in scene
  - Activate correct shader program
  - Bind vertex array object
  - Configure OpenGL render settings
  - Send values to be used in uniform variables
  - Call ***glDrawArrays*** function

Object3D Classes:
Scene, Group, Camera, Mesh

Object3D

- Object3D: scene graph nodes, represent objects in world
  - A matrix to store its local transformation
  - A list of reference to child objects
  - A reference to a parent object
- Functions:
  - **Add** and **remove** to update parent and child references
  - **getWorldMatrix** to calculate the world transformation
  - **getDescendantList** to convert tree to list
  - Transformation functions (translate, rotate, and scale)

Extending Classes: Scene and Group

- **Scene** class represents root node of scene graph tree
- **Group** class represents interior node

- These classes do not add functionality
  - But make application code easier to read

Extending Classes: Camera

- **Camera** class represents virtual camera used to view the scene
  - Previously, we assume the camera looks at the negative Z direction, but in reality, the camera could look at any direction
- Camera transform affects apparent placement of the objects in rendered image of the scene
  - E.g., camera shifts to left vs. scene objects shift to right
  - Each camera transformation affects scene objects in the opposite way
  - Define *view matrix* as inverse of camera transform matrix

Extending Class: Mesh

- **Mesh** class stores
  - Transformation matrix
  - Vertex Array Object
  - Geometry and Material instance
  - Visibility (Boolean) variable
    - Degerming if an object is visible or not.
- Tasks
  - Set up the association between vertex buffers (geometry) and shader variables (material)